Machine Learning for Spectral Analysis

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1. Introduction

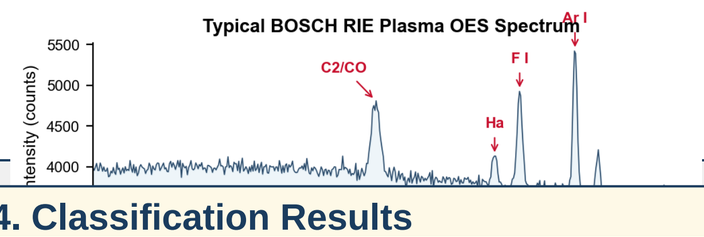
Optical Emission Spectroscopy (OES) provides **non-invasive, real-time diagnostics** of plasma processes by analysing emission spectra from excited species. Plasma-based manufacturing (semiconductor etching, surface treatment, thin-film deposition) relies on OES for process monitoring, but manual spectral interpretation is slow and error-prone.

This project develops an **automated ML pipeline** for plasma OES analysis targeting four objectives:

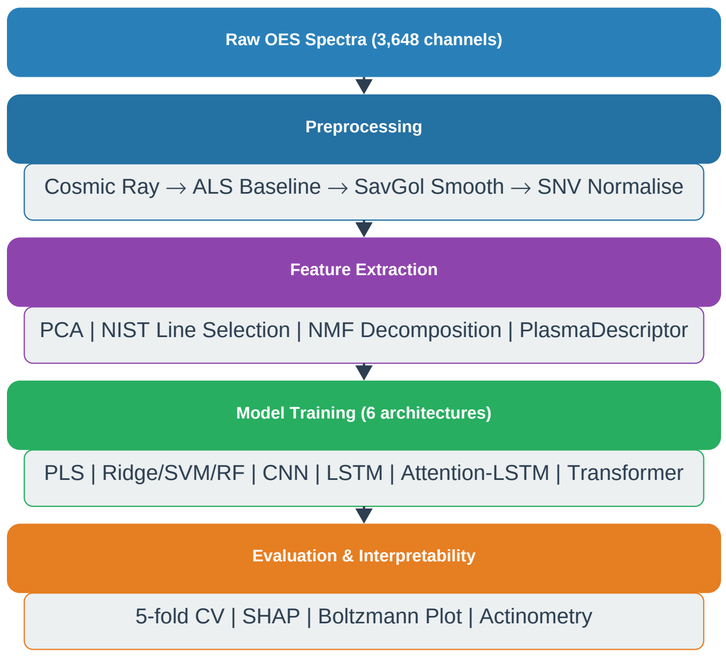
1. **Spectral feature identification** — automated peak detection, NMF decomposition, NIST database matching
2. **Plasma species classification** — SVM, RF, CNN, Transformer comparison (6 models)
3. **Spatiotemporal evolution** — Attention-LSTM phase prediction, species time-series extraction
4. **Semi-quantitative intensity** — actinometry (Coburn & Chen 1980), Boltzmann Te estimation

Three public datasets:

Dataset	Channels	Notes
LIBS Benchmark	40,002 ch	Steel composition (12 classes)
Mesbah CAP	51 ch	N2 plasma T_rot / T_vib
BOSCH RIE	3,648 ch	25 Hz, 10 days, SF6/C4F8



2. Methodology



Hyperparameter optimisation: Optuna two-stage search (20 trials per target)

Key Design Decisions

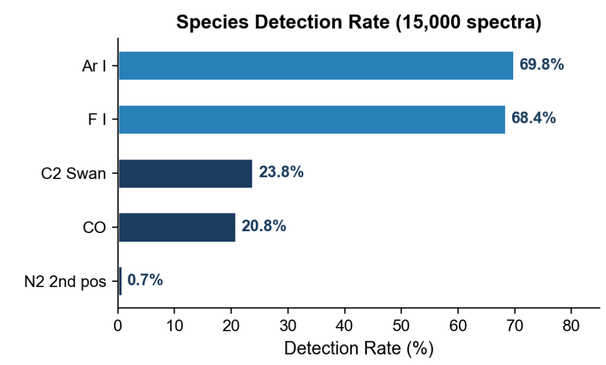
- **Per-element routing:** Cr uses Ridge+PCA; others use ANN+NIST.
- **GroupKFold CV:** Prevents same-target leakage across folds.
- **Balanced weights:** Compensates 12.1% minority class (OFF).
- **NMF over PCA:** Non-negative = physically interpretable spectra.

Preprocessing: Cosmic ray removal (5σ Z-score) → ALS baseline → SavGol smoothing → SNV normalisation. Average SNR gain: **10.99 dB** (12.6x improvement).

3. Species Identification

Non-negative Matrix Factorization (NMF) decomposes the spectral matrix **X** ≈ **W · H**, where each row of **H** is a pure-species emission spectrum and **W** contains the corresponding concentrations. NMF is physically appropriate because emission intensities are inherently non-negative.

NIST matching: 39 reference lines, 13 species, +/-1.5 nm tolerance. Detection threshold: peak > mean + 3 std.

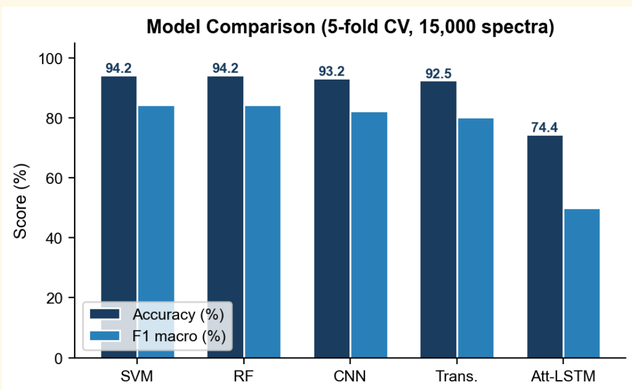


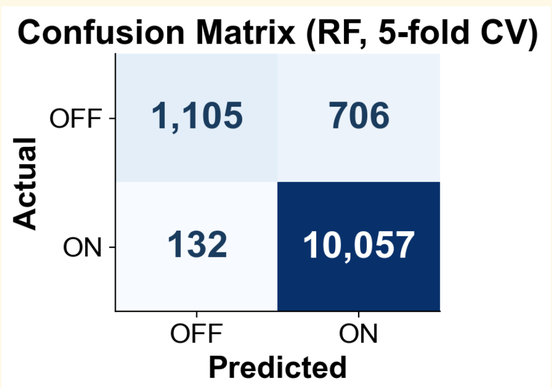
Species	Lines (nm)	Origin
F I	685.6, 690.2, 703.7	SF6 etchant radical
Ar I	696.5, 750.4, 763.5	Carrier gas reference
C2 Swan	473.7, 516.5, 563.6	C4F8 carbon indicator
CO	451.1, 519.8, 561.0	C4F8 + O2 product
Si I	250.7, 252.4, 288.2	Si etch product
SiF	440.0, 442.5	Si + F recombination

Validation: NMF Component 0 peaks at 684.4 nm (≈ F I 685.6); Component 2 at 515.1 nm (approx. C2 Swan 516.5) — unsupervised decomposition confirms NIST species independently.

4. Classification Results

94.2% Accuracy (SVM/RF, 5-fold CV)





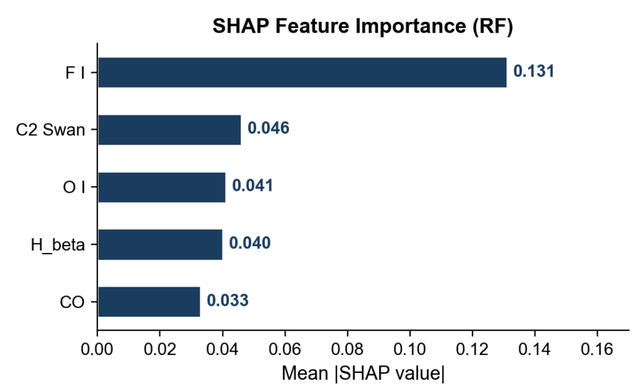
Species Detection (13 species):

Species	Rate	Species	Rate
Ar I	69.8%	C2 Swan	23.8%
F I	68.4%	CO	20.8%

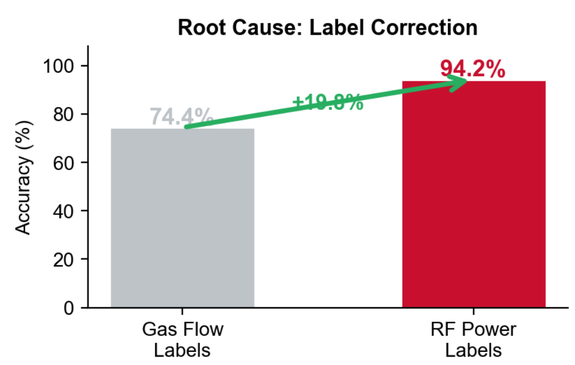
Model Architectures:

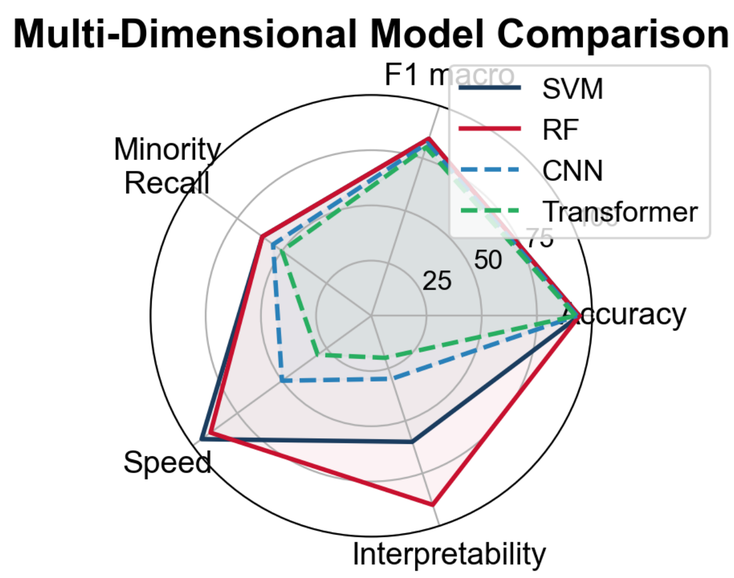
- **SVM/RF:** StandardScaler + RBF kernel (C=10) / 200 trees. Balanced weights.
- **CNN:** Conv1D(32-64-128) + AdaptiveAvgPool + FC(64) + Dropout(0.3).
- **Transformer:** 1D patch(64), d=128, 4 heads, 3 layers. [CLS] + AdamW.
- **Att-LSTM:** 2-layer LSTM(64) + additive attention. PCA(20) windows.

5. Interpretability & Physics



F I (fluorine radical) = most discriminative species, consistent with SF6 etchant chemistry.





Boltzmann T_exc = 13,334 K (6 Ar I lines).
DTW K-Means (k=4): 4 discharge phases

6. Conclusions & Further Work

Key Achievements:

✓ 13 species detected, 39 NIST lines, NMF validated
✓ 94.2% classification (6 models compared)
✓ SHAP: F I = 0.131 (physically validated etchant)
✓ Label correction: 74% → 94% via root-cause analysis
✓ T_rot = 20.0 K, T_vib = 102.0 K (CAP regression)
✓ 78 tests, 6 CLI modes, fully reproducible

Limitations:

- OES→process parameter regression failed (R2 < 0, causal irreversibility)
- Boltzmann Te limited by narrow Ar I energy spread (0.33 eV)
- Spatial etch prediction lacks wafer ID alignment

Further Work:

- Extend to multi-class species classification (beyond binary ON/OFF)
- Implement real-time OES monitoring with streaming inference
- Apply transfer learning across different plasma reactors

Project Metrics: 15,000 spectra analysed | 3,648 spectral channels | 13 species identified | 39 NIST emission lines | 6 ML models compared | 78 automated tests | 32 development stories | 23 literature references | 6 CLI task modes | 3 public datasets

References:

[1] Gidon *et al.* (2019) IEEE Trans. Radiat. Plasma Med. Sci.